

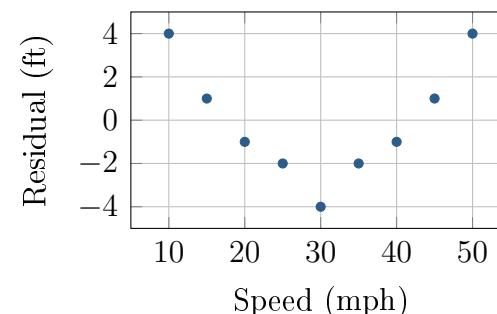
One Line, Two Model Diagnostics

Unit 5 · Topic 5.4 · THE PROBLEM

Suppose a road-safety lab records braking distance y at speeds x from 10 through 50 mph on one vehicle and dry surface. The line is $\hat{y} = -10 + 2.5x$, the correlation is $r = 0.997$, and the residual plot is shown. A residual is actual distance minus the line's prediction. At 50 mph, the actual distance is 119 feet.

Ezra: The large correlation and residuals on both sides of 0 justify using the line across the tested speeds.

June: I would use where the residuals occur and the 50-mph error to judge safety, then check any replacement model separately.



FIRST TAKE — one sentence before you work the parts

Would you trust this line to plan a safe stopping distance? Give one number or visible pattern in one sentence.

- DOK 1** (a) Read the residual at 50 mph. Does the line predict too high or too low there?
- DOK 2** (b) Verify the 50-mph residual using the equation. Describe where the residual plot is above and below zero.
- DOK 3** (c) Whose judgment is supported? Explain why the residual pattern matters even with the large correlation, and connect the 50-mph error to safety. State what you would check in a replacement model for this vehicle and dry surface. Write 3–4 sentences.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.